State Estimation and Filtering for Robotic Perception

AA273

Optimization based estimation



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 5 due Fri Feb 20
- HW 6 out Fri Feb 20, due Fri Mar 6
- Project Proposal: due Tue Feb 17
  - 2-3 pages, Latex
  - Related work >= 10 refs, ~1 page
  - Proposed work ~ 1 page
  - Bibtex refs

Lecture 13:

- Optimization based estimation
- Laplace's approximation
- Factoring trajectories

- Project Reports: due Tue Mar 17
  - 5-6 pages + 1 page per group member > 1
  - Latex/Overleaf, IEEE conf format
  - >= 10 refs
  - Describe research and results
  - Re-use proposal material
- Project Presentations: 3/10, 3/12
  - 5 min pres + 3 min Q and A
  - Sign up for preferred time slot
  - Share slides through zoom meeting
  - Present on your computer

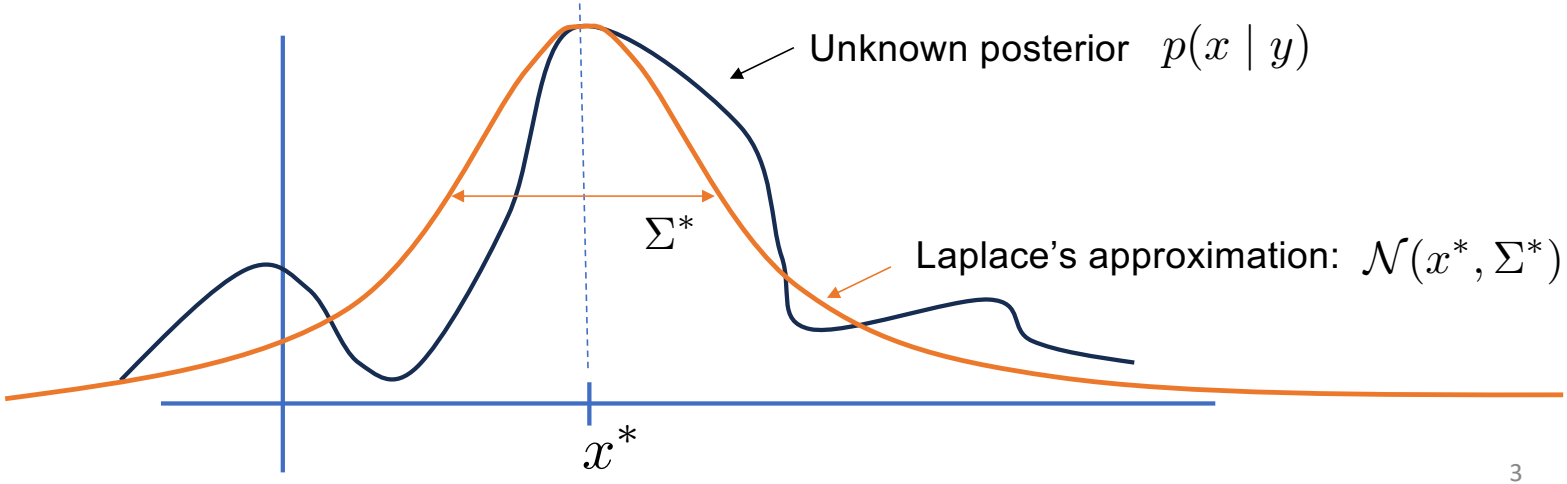
Optimization based estimation

Maximum A Posteriori (MAP) Estimation

Suppose we can't find posterior directly: $p(x \mid y)$

Can we find max of the posterior?: $x^* = \arg\max_x p(x \mid y)$

Even better, can we find a **Gaussian approximation** with mean at x*?:



Gaussian Special Case

Consider a known Gaussian posterior (just to build intuition):

$$p(x \mid y) = \eta e^{-\frac{1}{2}(x-\mu_{x|y})^T \Sigma_{x|y}^{-1}(x-\mu_{x|y})}$$

Mahalanobis distance notation:

$$\|x - \mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}} = (x - \mu_{x|y})^T \Sigma^{-1}_{x|y}(x - \mu_{x|y})$$

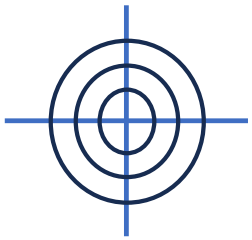
Rewrite Gaussian as:

$$p(x \mid y) = \eta e^{-\frac{1}{2}\|x-\mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}}}$$

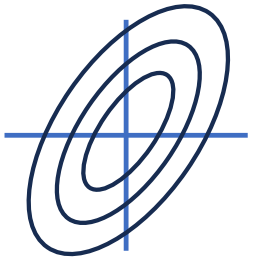
Let's solve:

$$x^* = \arg\max_x p(x \mid y)$$

ℓ_2 distance



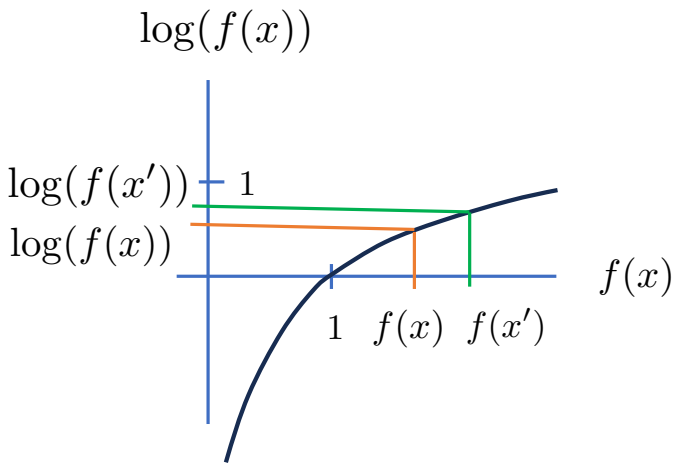
Mahalanobis distance



Negative log trick

The following is true for any positive-valued function:

$$\arg\max_x f(x) = \arg\max_x \log f(x)$$



Because log(.) is monotonically increasing:

$$f(x) \leq f(x') \iff \log(f(x)) \leq \log(f(x'))$$

Also we can convert to a min problem:

$$\arg\max_x f(x) = \arg\min_x -f(x)$$

Combining there, let's solve:

$$x^* = \arg\max_x p(x \mid y) = \arg\min_x -\log(p(x \mid y))$$

Solve Optimization

$$x^* = \arg\min_x -\log(p(x \mid y))$$

$$= \arg\min_x \left(-\log(\eta) + \|x - \mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}} \right)$$

Constant doesn't affect argmin

$$= \arg\min_x \frac{1}{2}\|x - \mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}}$$

Convex quadratic, solve analytically

$$\nabla_x \frac{1}{2}\|x - \mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}} = \Sigma^{-1}_{x|y}(x - \mu_{x|y})$$

$$\Sigma^{-1}_{x|y}(x^* - \mu_{x|y}) = 0$$

$$(x^* - \mu_{x|y}) = 0 \implies \boxed{x^* = \mu_{x|y}}$$

Assuming $\Sigma^{-1}_{x|y}$ full rank

Find Covariance from Hessian

$$x^* = \arg\min_x -\log(p(x \mid y))$$

$$\nabla_x \frac{1}{2}\|x - \mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}} = \Sigma^{-1}_{x|y}(x - \mu_{x|y})$$

$$\nabla^2_x \frac{1}{2}\|x - \mu_{x|y}\|^2_{\Sigma^{-1}_{x|y}} = \nabla_x \Sigma^{-1}_{x|y}(x - \mu_{x|y}) = \Sigma^{-1}_{x|y}$$

Claim: the inverse of the Hessian gives the covariance of Laplace's Approx

$$\Sigma^* = (\nabla^2_x p(x^* \mid y))^{-1}$$

Laplace's Approx: $p(x \mid y) \approx \mathcal{N}(x^*, \Sigma^*)$

Graphics on board, in notes

Consider Bayes Rule with Linear-Gaussian model

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)}$$

$$\arg\max_x p(x \mid y) = \arg\min_x \left(-\log p(y \mid x) - \log p(x) + \log p(y)\right)$$

$$\arg\max_x p(x \mid y) = \arg\min_x \left(-\log p(y \mid x) - \log p(x)\right)$$

Consider Linear-Gaussian model:

Likelihood:

$$Y = CX + V \qquad V \sim \mathcal{N}(0, R)$$

$$-\log(p(y \mid x)) \propto \frac{1}{2}\|y - Cx\|^2_{R^{-1}}$$

Prior:

$$X \sim \mathcal{N}(\mu_x, \Sigma_x)$$

$$-\log(p(x)) \propto \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$$

Linear-Gaussian Case

We have:

$$x^* = \arg\min_x \left(\frac{1}{2}\|y - Cx\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}} \right)$$

Again, convex quadratic optimization.

Claim (you prove in PS5): The solution is given by the the mean from the Kalman Filter update step! The Laplace approx. is given by both mean and covariance from KF update step.

$$x^* = \mu_x + K(y - Cx)$$

$$\Sigma^* = \Sigma_x - KC\Sigma_x$$

$$K = \Sigma_x C^T (C\Sigma_x C^T + R)^{-1}$$

Gauss-Newton Algorithm

On the board and in the notes.

Nonlinear Case

On the board and in the notes.

Gauss-Newton Algorithm and EKF

On the board and in the notes.